The Hong Kong Polytechnic University

Department of Computing

COMP4913 Capstone Project

Final Report

Food Component Detection for Dietary Recommendation- Children's Fun Nutrition Guide

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1. Abstract

This article introduces the design, implementation and evaluation of the "Children's Fun Nutrition Guide". The platform is a web-based platform aimed at improving the eating habits and nutritional literacy of children aged 3 to 12. In view of the increasing attention to childhood obesity recorded by the **World Health Organization (WHO)** and the U.S. Centers for **Disease Control and Prevention (CDC)** and its developmental risks associated with poor diet, the platform adopts a child-centered holistic approach, integrating artificial intelligence, interactive design and evidence-based nutrition guidelines.

The system is deployed on kids-nutrition-fun.com, based on the WordPress architecture hosted by **SiteGround**, and equipped with custom PHP and JavaScript extensions. Its main functions include: Nutri, a bilingual (English/Chinese) artificial intelligence nutrition assistant driven by the **Azure OpenAI** gateway (**GPT-4o Mini**) of the Department of Computer Science of the Hong Kong Polytechnic University; a recipe recommendation system containing more than 50 child-friendly recipes, covering seven meal categories; and two browser-based educational games ("Nutrition Parkour " and "Nutrition Plate Builder"). And a user registration system that supports personalized multi-child file management.

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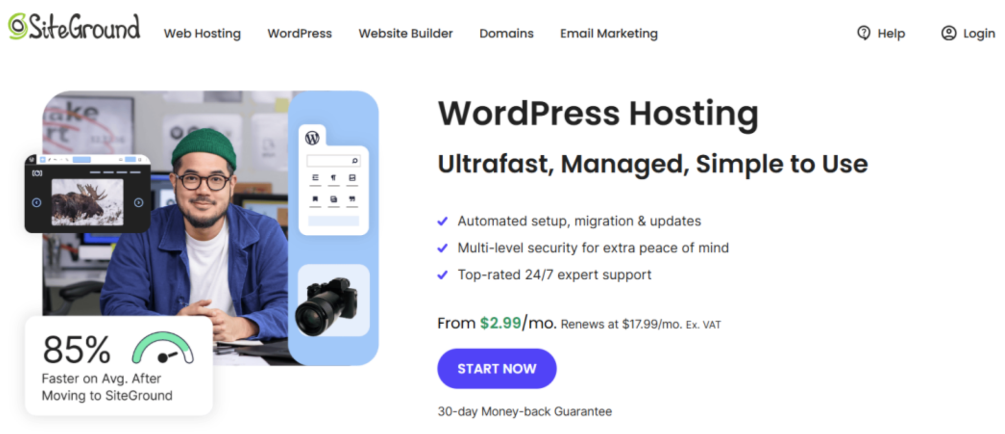
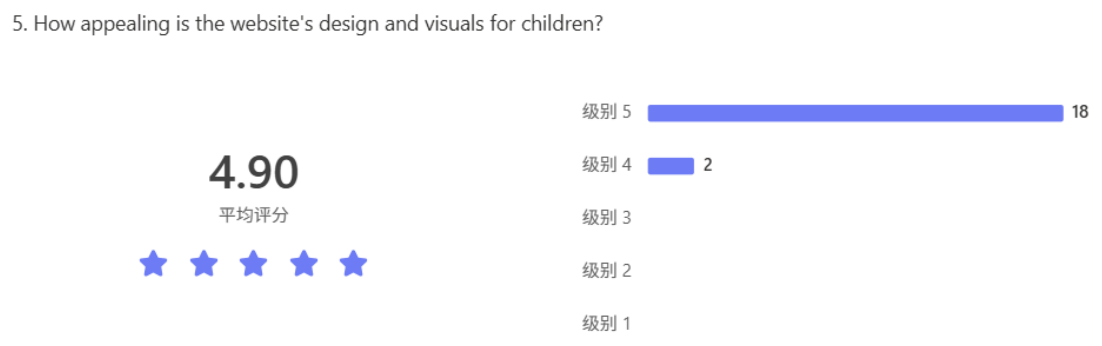
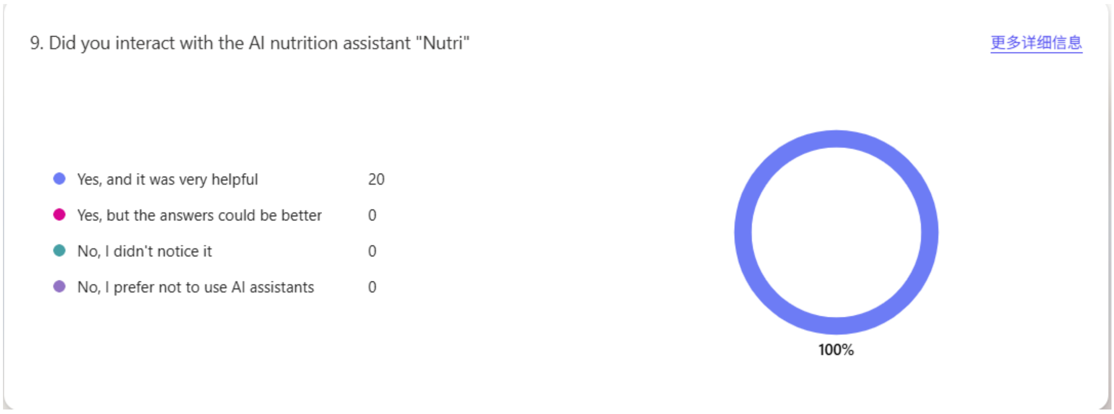


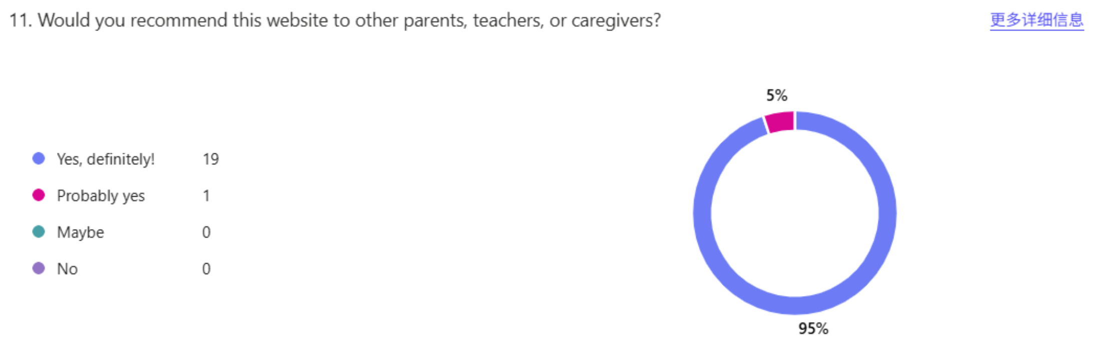
All nutrients and recommendations comply with the dietary guidelines of the World Health Organization (WHO) and the U.S. Centers for Disease Control and Prevention (CDC), and the platform adopts role-based access control to ensure the privacy and security of children's data.

We evaluated the usability of **20** participants (including parents, teachers, and caregivers). The results show that user acceptance is very high: **95%** of the participants expressed their willingness to recommend the platform to others, **100%** of the participants found the nutritional assistant very helpful, and the average score of visual design was **4.9 (out of 5)**. During the test, the artificial intelligence assistant successfully handled more than **95%** of nutrition-related inquiries. These results show that gamification and artificial intelligence-assisted personalization can effectively bridge the gap between clinical dietary guidelines and practical applications at home. Future work will explore image-based food records and vertical behavior research to further enhance the influence of the platform. Evaluation form: https://forms.cloud.microsoft/r/HBigxkE7s3

Keywords: Children's Nutrition; AI chatbot; Recipe Recommendation; Gamification; Web Application







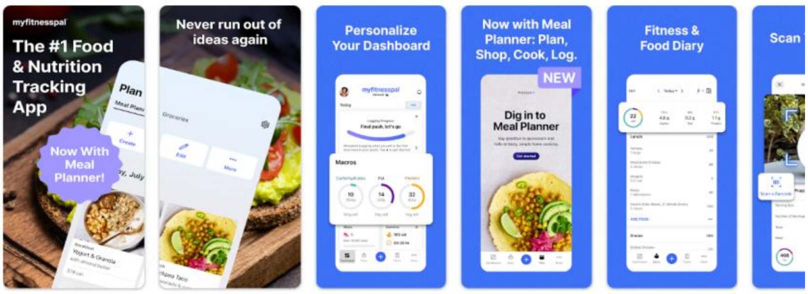
2. Main body of text

Chapter 1: Introduction

1.1 Project Background

3 to 12 years old is a critical period to develop lifelong eating habits. The World Health Organization **(WHO)** and the U.S. Centers for Disease Control and Prevention **(CDC)** pointed out that malnutrition in early childhood is closely related to growth retardation, obesity, decreased immunity, and increased risk of chronic diseases such as type 2 diabetes and cardiovascular disease [1] [2]. Although this has been fully proven, many parents still face many challenges in developing healthy eating habits for their children, especially when faced with picky children, contradictory dietary information, and lack of interactive tools suitable for children's age.

At present, the digital health field provides rich nutrition and fitness applications, such as MyFitnessPal and Cronometer, but these platforms are mainly aimed at adult users. They rely on the calorie calculation interface, digital dashboard and a large amount of text content, which are not suitable for young children. There is an urgent need for a platform to combine evidence-based dietary guidance with child-friendly interactive design, so as to inspire children to regard healthy eating as a pleasure rather than a burden.



This project aims to fill this gap by developing the "Children's Fun Nutrition Guide" network platform at kids-nutrition-fun.com. The platform integrates artificial intelligence nutrition assistant, personalized recipe recommendation system, interactive educational games and parent-managed user registration system. All functions run in a safe and reliable children's environment.

1.2 Project Objectives

The main goal of this project is to design, implement and evaluate a comprehensive network platform to provide personalized nutrition guidance and interactive learning experiences for children aged 3 to 12 and their parents. To this end, this project aims to achieve the following functional and non-functional goals:

Functional objectives

- Child-friendly interface: Design an fascinating user interface that meets the characteristics of children's age, integrating cartoon themes, large interactive elements and intuitive navigation, suitable for children aged 3 to 12.
- AI Nutrition Assistant: Implement Nutri, a bilingual (English/Chinese) AI chatbot based on the **COMP Azure OpenAI** gateway of the Hong Kong Polytechnic University, which can answer children's nutrition questions in a safe and age-appropriate manner.
- Recipe recommendation system: Develop a browsable recipe catalog, including more than 50 recipes suitable for children, covering seven meal categories,

complying with the dietary guidelines of the World Health Organization and the U.S. Centers for Disease Control and Prevention, and can be screened according to health goals and dietary preferences.

- Educational mini-games: integrate interactive browser games - "Nutrition Parkour " and "Nutrition Plate Builder" - to strengthen children's nutritional literacy through game-based learning.
- User registration and personal information management: Parents are allowed to register accounts and manage multiple children's personalized information, including age, allergy history, dietary preferences, and health goals.

Non-functional objectives

- Performance: The page loading time is less than 3 seconds, supports at least 10 concurrent users, and the performance does not decrease.
- Ease of use: Ensure easy access to WCAG-standard, touch-friendly and responsive interfaces on desktops, tablets, and mobile devices.
- Security and privacy: Enforce HTTPS encryption, role-based access control, and strict data privacy measures to protect children's personal information.

1.3 Project Significance and Contributions

This project has made many contributions to the field of pediatric digital health:

- Child-centered AI integration: By embedding bilingual conversational AI assistants specifically for children's nutrition, this project demonstrates the practical and safety applications of large-scale language models in the field of pediatric health - which is rarely paid attention to in existing literature.
- Gamification of nutrition education: By integrating interactive games, this project provides an evidence-based method to encourage children to develop a positive eating attitude and solve the problem of picky eating through education and fun, rather than rigid guidance.
- Personalized recommendations based on evidence-based medicine: All recipe recommendations and diet contents are based on the authoritative guidelines of the World Health Organization (WHO) and the U.S. Centers for Disease Control and Prevention (CDC) to ensure scientific rigor and take into account children's understanding.

- Scalable open architecture: The WordPress-based platform provides a scalable foundation for future feature enhancements, such as recording food images through computer vision, multilingual support, and integration with wearable health devices.

1.4 Report Organization

The rest of this report is structured as follows:

- The second chapter reviews the relevant literature on children's nutrition challenges, existing digital health tools, the application of artificial intelligence in health applications, and game-based education.
- The third chapter describes the system design and architecture, including three-layer Web architecture, front-end UI/UX design, back-end infrastructure, database mode and artificial intelligence integration.
- Chapter 4 records the realization of all core functions, including nutrition artificial intelligence assistant, recipe system, small games and user management.
- Chapter 5 discusses the technical and design challenges encountered in the development process and the solutions adopted.
- Chapter 6 introduces the test methods, functional test results and user evaluation results.
- Chapter 7 summarizes the results of the report, the limitations of the project and the future direction of work.

Chapter 2: Literature Review

2.1 Childhood Nutrition Challenges

Adequate nutrition in early childhood (3-12 years old) is widely regarded as a key factor affecting physical growth, cognitive development and long-term health outcomes. The World Health Organization **(WHO)** pointed out that an unhealthy diet is one of the main risk factors for global non-communicable diseases, while the U.S. Centers for Disease Control and Prevention **(CDC)** reports that poor eating habits developed in childhood can significantly increase the risk of obesity and metabolic disorders in adulthood [1] [2]. The systematic review of Kozioł-Kozakowska (2023) further emphasizes that nutritional deficiency in early childhood - especially iron, vitamin D and calcium deficiency - is associated with impaired neurodevelopment and weakened immune function [3]. Despite this evidence, picky eaters, parents' lack of nutritional knowledge, and a large amount of contradictory dietary information remain the main obstacles to children's access to optimal nutrition.

2.2 Existing Digital Nutrition Tools

The digital health market provides a wide variety of nutrition tracking applications, such as MyFitnessPal, Cronometer, and Yazio, which have been widely used by adult users [4]. These tools usually provide calorie calculation, macronutrient tracking, and food diary functions. However, they have a common limitation: their interfaces, terms, and interaction patterns are specially designed for adult users and are not suitable for the cognitive and behavioral characteristics of young children. At present, there is no mainstream business platform that can provide comprehensive, child-centered nutritional tools, combining personalized dietary advice with an interactive mechanism suitable for children's age. This market gap is both a design challenge and an opportunity for innovation in the field of pediatric digital health.

2.3 Artificial Intelligence in Health Applications

Large Language Model (LLM) has great potential in the field of health care, including patient education, symptom triage, and personalized dietary advice [5].

A conversational artificial intelligence system based on models such as GPT-4 has been proven to provide nutritionally accurate and contextual responses to diet-related problems. In some evaluation tasks, its performance is comparable to that of human nutritionists [6]. However, deploying LLM in a pediatric environment will bring specific security and content review problems: replies must be limited to the scope of topics suitable for children's age, avoid harmful content, and use easy-to-understand language. This project meets these requirements through the carefully designed system prompt architecture applied to the PolyU COMP Azure OpenAI gateway, and limits the function of Nutri Assistant to the theme of children's nutrition.

2.4 Gamification in Education

Gamification - the application of game design elements to non-game situations - has been widely studied as a teaching strategy to improve learner participation and knowledge retention [7]. In the field of health education, game-based intervention has been shown to have a positive impact on the eating behavior and nutrition awareness of school-age children, especially when the feedback is timely, and the reward has an inherent motivation effect [8]. For example, the U.S. Department of Agriculture's (USDA) "Team Nutrition" interactive game has been confirmed to significantly improve children's awareness of food categories and preference for healthy food for children aged 6-12 [9]. Based on this, this project integrates two browser-based educational games - Nutrition Parkour and the U.S. Department of Agriculture's "MyPlate" cultural breakfast game - aiming to strengthen nutritional literacy through interactive games.

Chapter 3: System Design and Architecture

3.1 Overall System Architecture

The platform adopts a classic three-layer Web architecture, including representation layer, application layer, and data layer, as shown in Figure 3.1.

The representation layer contains all client-facing pages rendered in the browser and is built using HTML5, CSS3, JavaScript, and Elementor visual page builders. The application layer is based on WordPress 6.x and PHP 8.x, and extends four key plug-ins: WooCommerce (recipe directory), Ultimate Member (user registration), a custom plug-in that handles Nutri chatbot API calls, and WPForms/ACF for forms and custom fields. The data layer stores all persistent content in the MySQL 8.x database, which is hosted in the SiteGround Singapore data center and enforces HTTPS/SSL encryption.

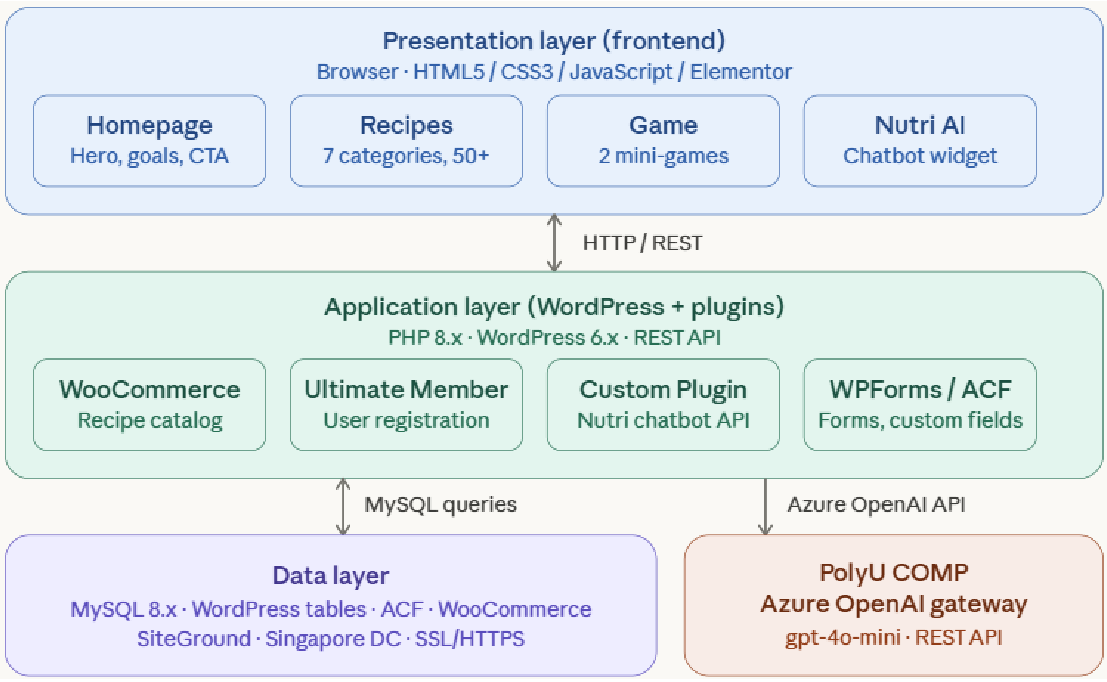


Fig. 3.1 — Three-tier system architecture of kids-nutrition-fun.com

3.2 Frontend Design

The user interface prioritizes children's participation while ensuring their trust in parents. Key principles include bright cartoon-style color schemes, easy-to-touch large buttons (minimum 44×44 pixels), sans-serif fonts (18-20 pixels Nunito/Quicksand fonts), and fully responsive layouts on desktops, tablets, and mobile devices. The content is divided into two styles: the information part (about us, contact us) is designed for parents in a professional and clear way, while the interactive part (recipes, games) uses lively visual language for children.

3.3 Backend Architecture and Plugin Stack

WordPress's plug-in ecosystem provides core back-end functions. WooCommerce is reused as a recipe management system - recipes are stored in the form of products without checkout functions, and the categories correspond to meal types (breakfast, lunch, dinner, snacks and desserts, salads and bowl dishes, soups and stews, vegetarian main dishes). Ultimate Member is responsible for parent account creation, email verification, and role-based access control. A custom WordPress plug-in manages all communications with the PolyU COMP Azure OpenAI gateway, including system prompt injection and token usage logging.

3.4 Nutri AI Chatbot Architecture

Nutri Assistant is embedded in the whole website in the form of a floating chat window. As shown in Figure 3.2, each user message will be intercepted by a custom plug-in, which will add a pre-set system prompt before the message, limit the reply to the scope of children's nutrition-related topics, and then forward the request to the PolyU COMP Azure OpenAI endpoint (https://comp.azure-api.net/azure) through API key header authentication. The deployed model is **gpt-4o-mini** (GPT-4o Mini, version 2024-07-18), which is chosen because it is cost-effective and only costs $0.00066 per 1000 tokens. The reply is returned in the user's language (English or Chinese) and presented directly in the chat window.

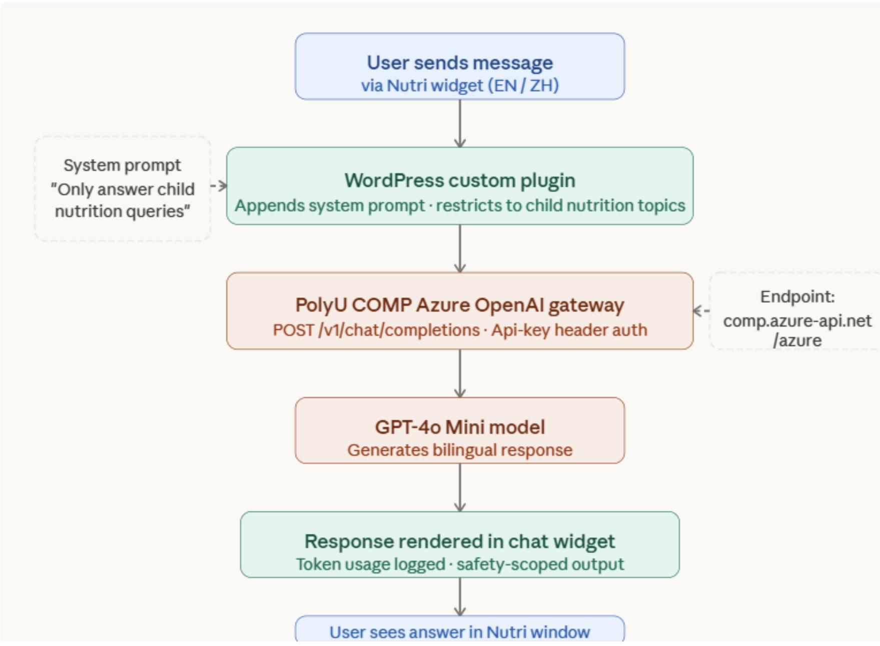


Fig.3.2 — Nutri AI chatbot request flow

3.5 Database Design

The database extends WordPress's native MySQL architecture and adds WooCommerce and ACF custom tables. Figure 3.3 shows the entity relationship diagram that reflects the actual system structure.

The platform uses a single unified user entity for account management and is implemented through the Ultimate Member plug-in. Each Registered User Can Record The recipe interaction through the Recipe_log bridge table, which records the recipes viewed or saved by the user (One-To-Many Relationship). Each recipe entry belongs to one or more Recipe_category records - corresponding to seven meal categories (Breakfast, Lunch, Dinner, Snacks and Desserts, Salads and Bowl Dishes, Soups and Stews, Vegetarian Main Dishes) - And may contain one or more Recipe_allergen entries for dietary screening. Nutritional fields, including calories, proteins, carbohydrates, and fats, are stored directly in each recipe record so they can be screened against nutritional standards in the future.

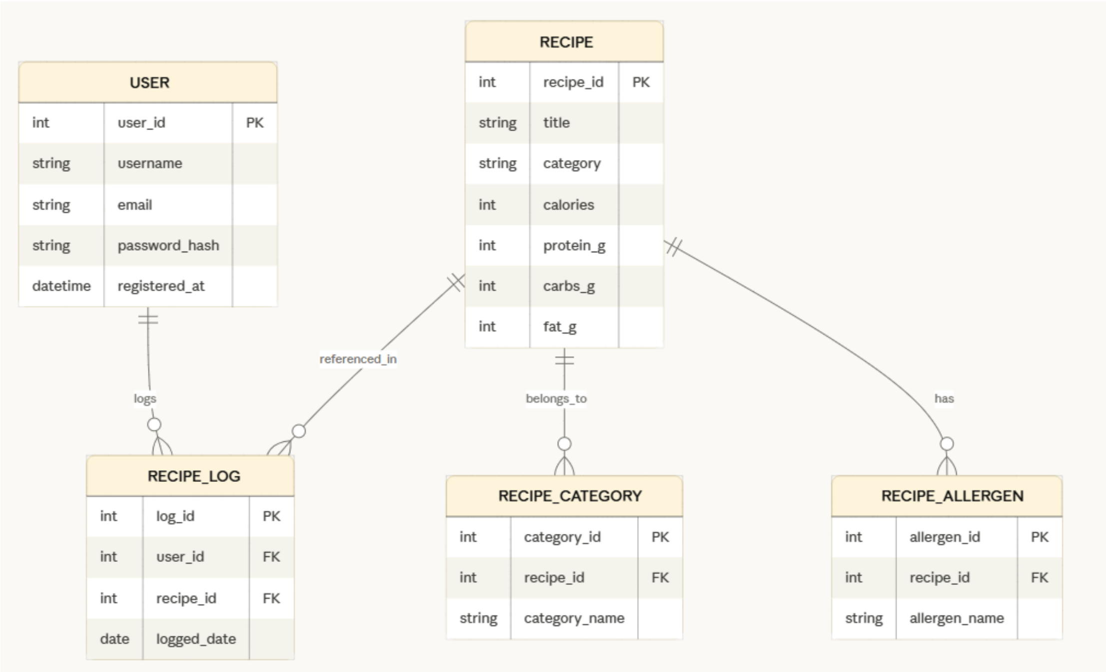


Fig. 3.3 — Database entity relationship diagram (ERD)

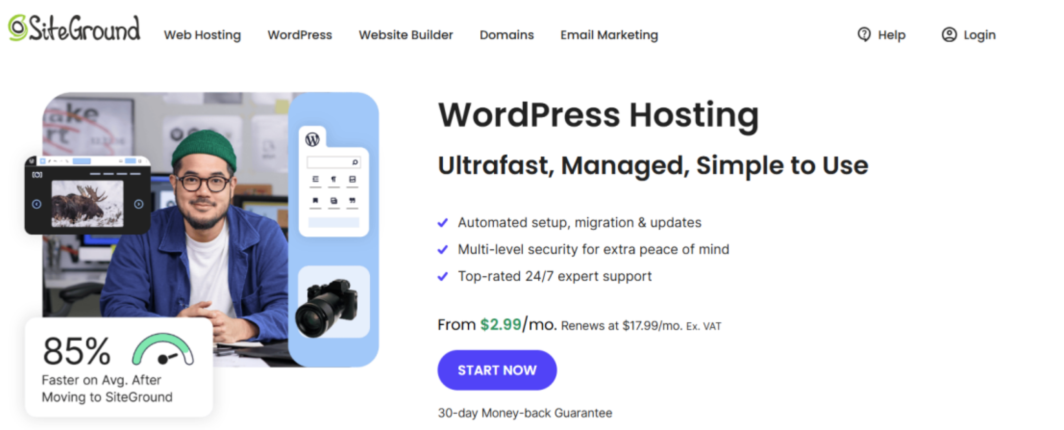
Chapter 4: Implementation Progress

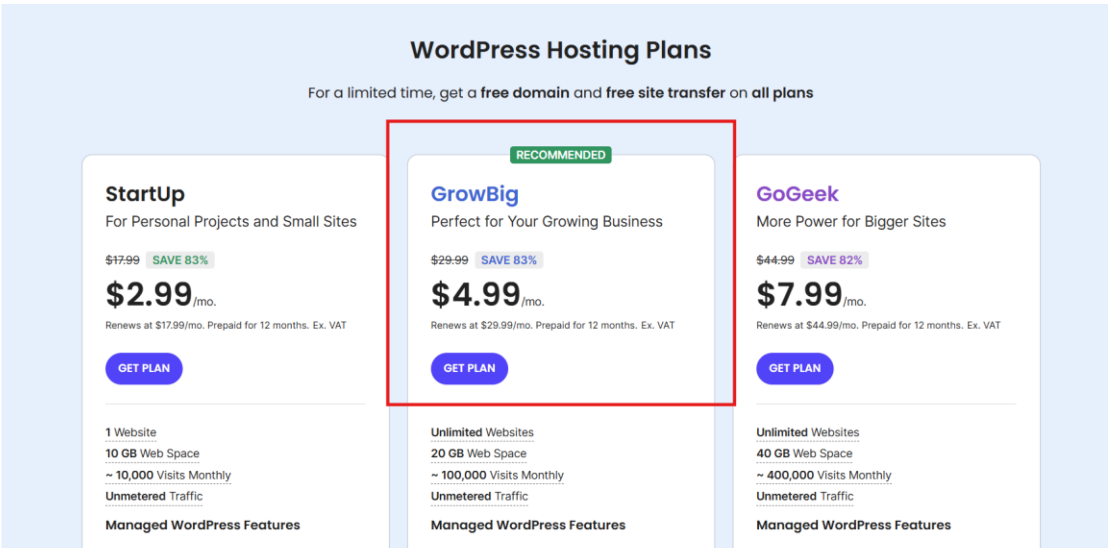
This chapter details the implementation work completed from December 2025 to early January 2026, and records the construction of technical infrastructure, core function development, plug-in integration, and continuous development activities.

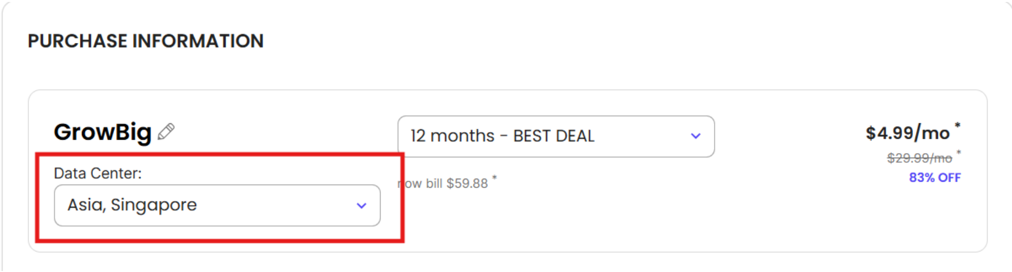
4.1 Technical Infrastructure Setup

4.1.1 Technical Infrastructure-Domain Acquisition and Setup

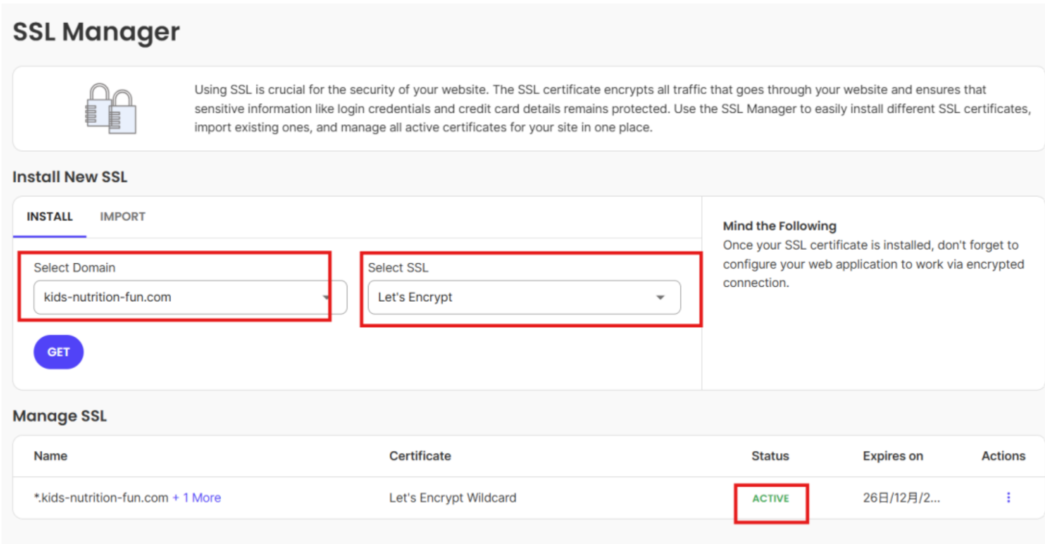
The platform is deployed on SiteGround's GrowBig shared hosting plan ($60 per year) and selects the Singapore data center to minimize the delay of the target Asian user base.

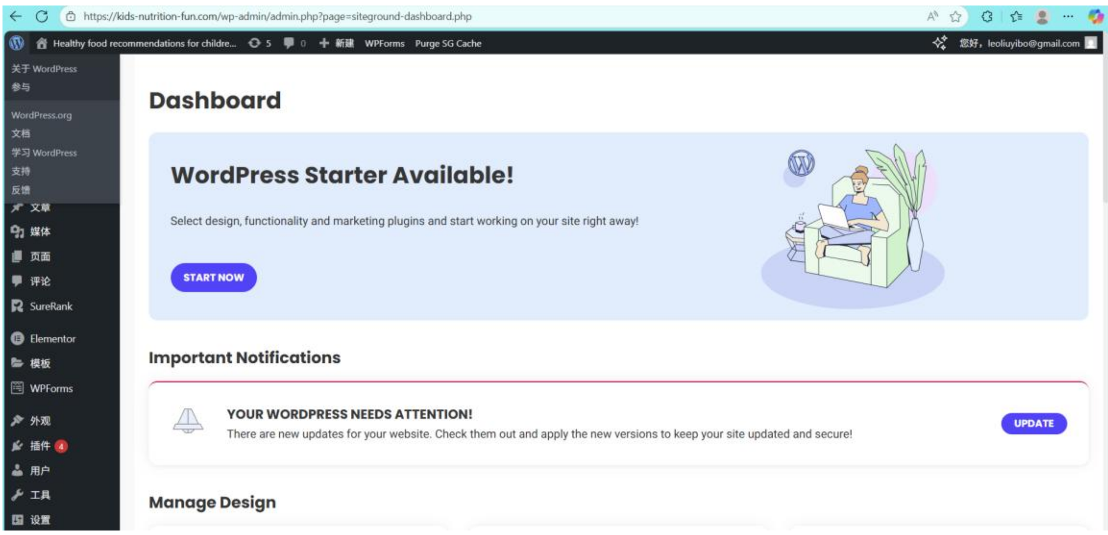






The domain name kids-nutrition-fun.com was registered in October 2025, and the DNS is directly managed by **SiteGround**. Let's Encrypt SSL certificate forces all pages to use HTTPS to ensure the encrypted transmission of user credentials and session data. Other infrastructure features include SiteGround's AI-driven firewall against robots and intrusions, as well as daily automatic backups for disaster recovery.





WordPress 6.x with PHP 8.x serves as the CMS and application runtime, connected to a

MySQL 8.x database that houses all content, user, and recipe data, as shown in Fig. 4.1.

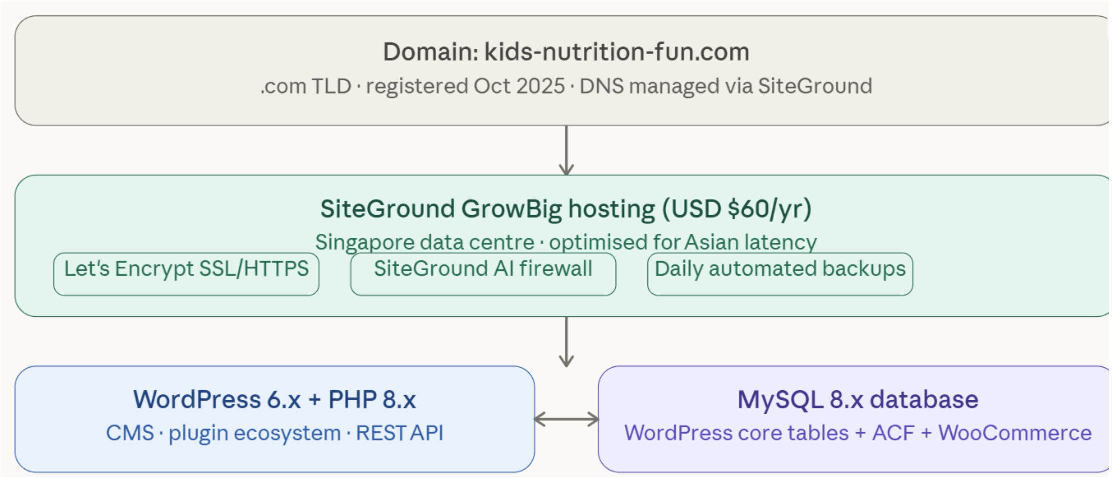


Fig. 4.1 — Hosting and infrastructure stack

4.2 Website Structure and Navigation

The website contains five main pages, which can be accessed through the fixed top navigation menu, as shown in Figure 4.2.

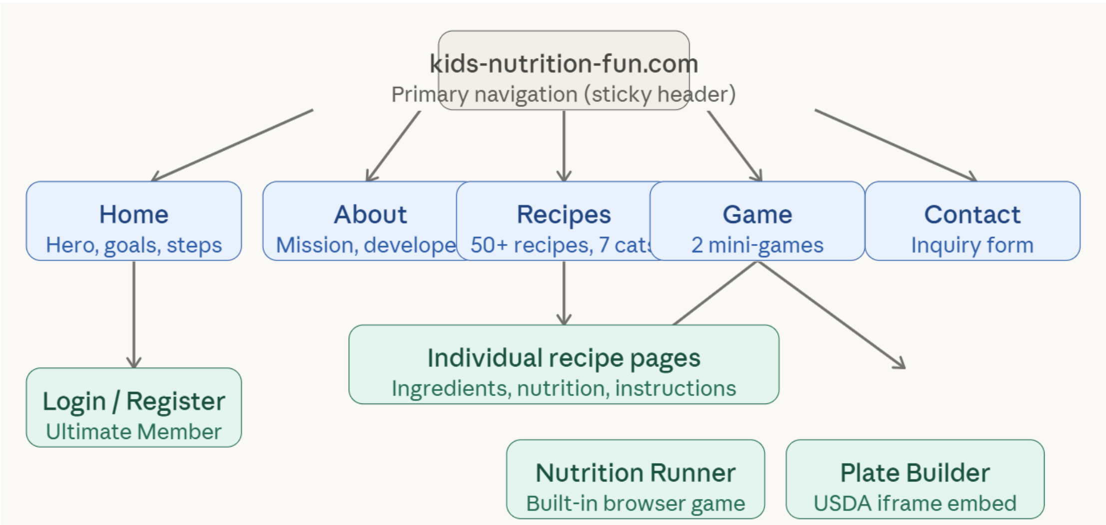
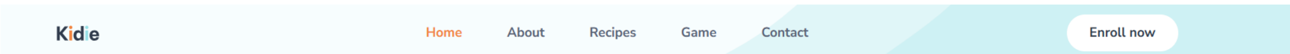
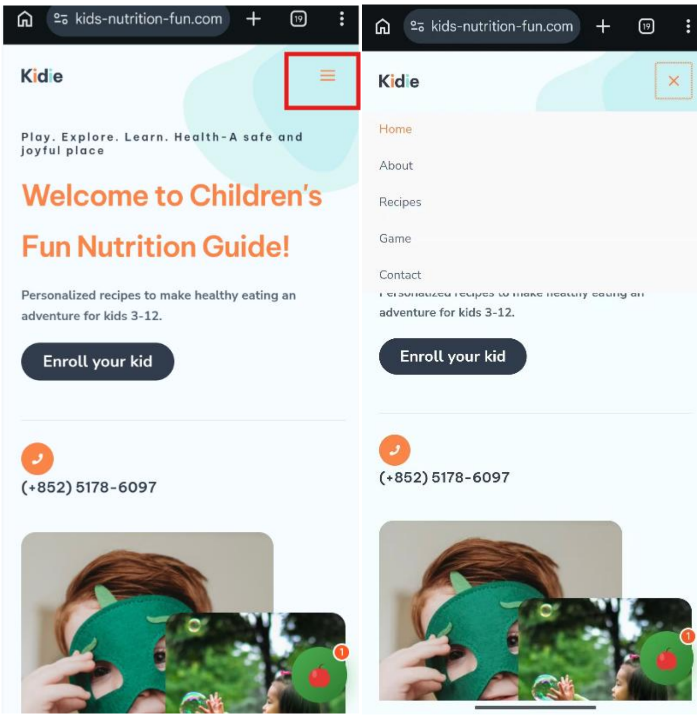


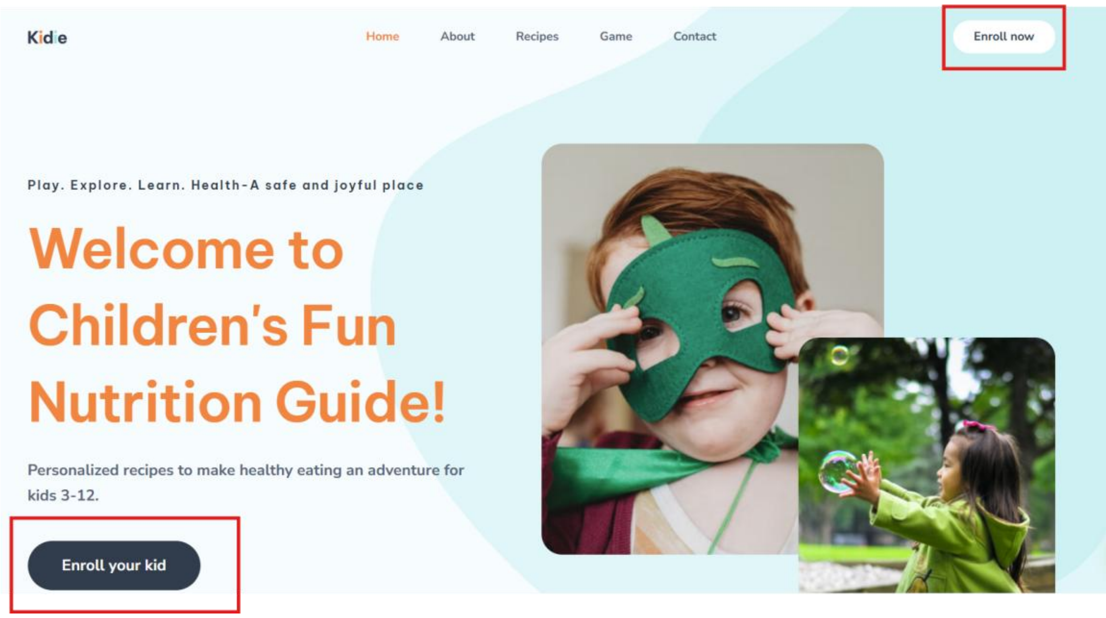
Fig. 4.2 — Site map and navigation structure



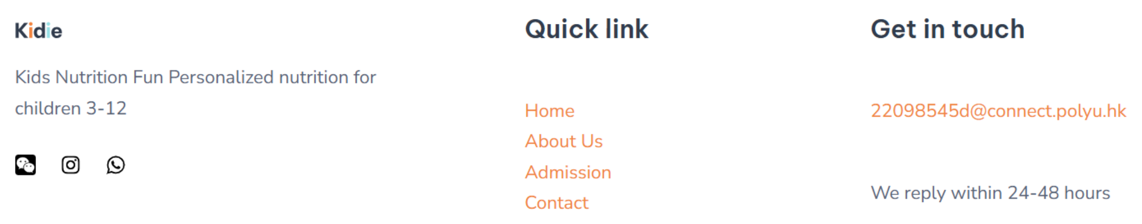
When scrolling the page, the header is always visible; when the screen size of the mobile device is less than 768 pixels, the header will be folded into a hamburger menu. All buttons meet the minimum click target size of 44×44 pixels specified by WCAG.



The "**Enroll Now**" button in the header will guide the user to the login/registration page.



The footer provides a quick link to the key page, as well as a pre-filled email contact link, and promises to reply within 24-48 hours.



4.3 Nutri AI Chatbot

Nutri AI Assistant is embedded throughout the website in the form of a floating chat window and implemented through custom WordPress plug-ins. As shown in Figure 4.3, each user message will be combined with pre-set system prompts and then sent to the PolyU COMP Azure OpenAI gateway.

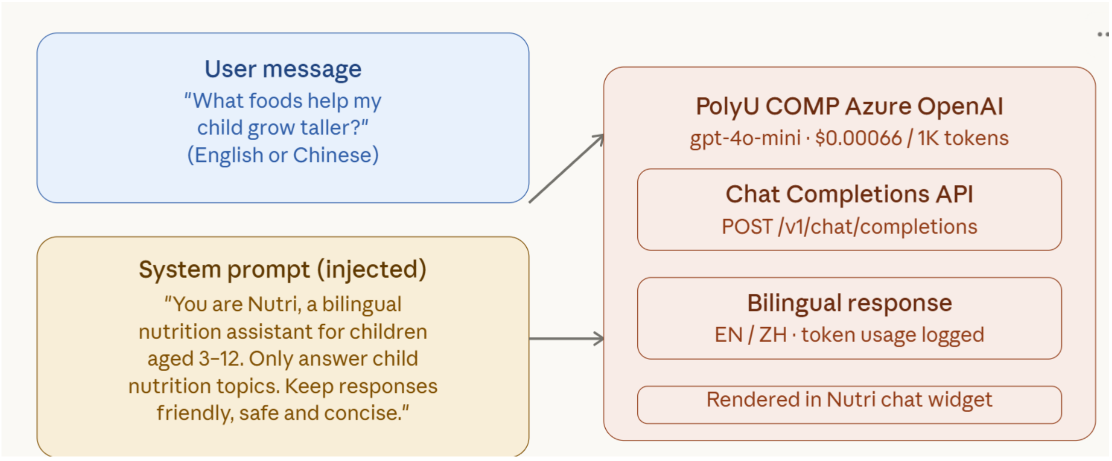
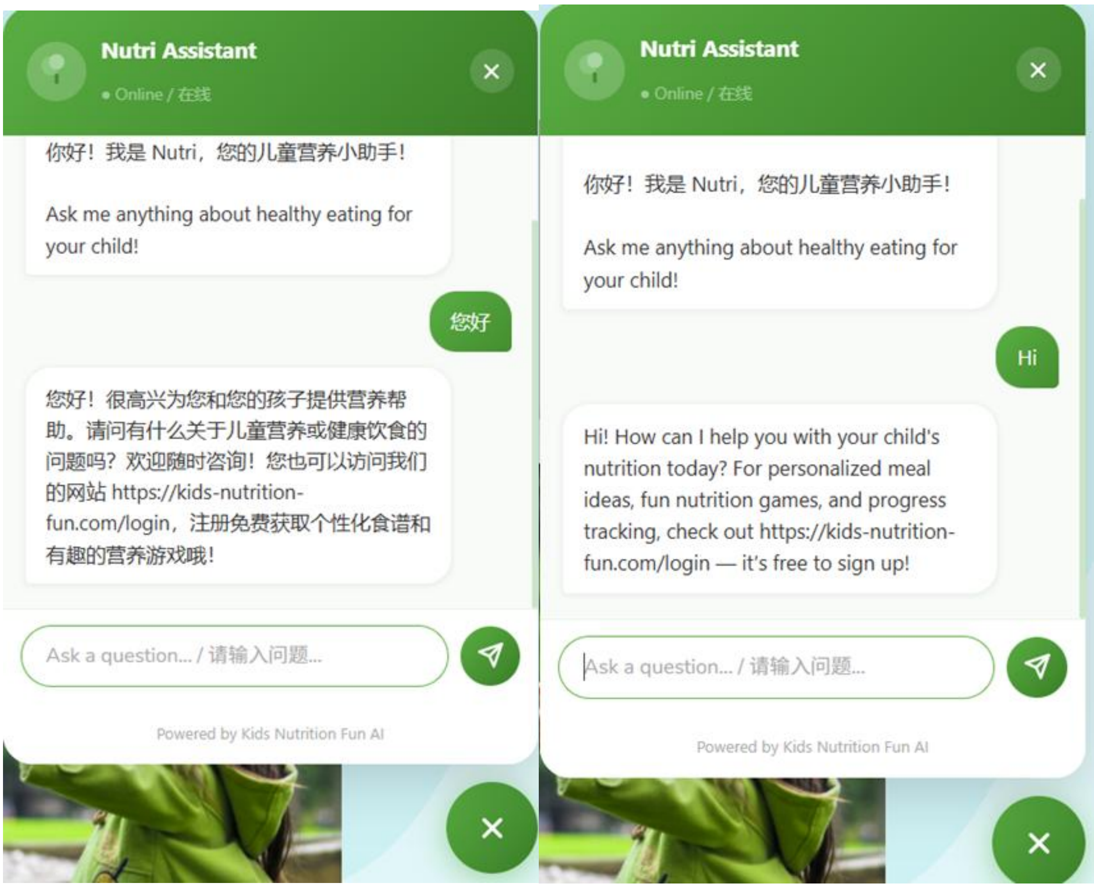


Fig. 4.3 — Nutri system prompt injection and API call design.

The system prompts the model to act as a bilingual (English/Chinese) nutrition assistant named Nutri, strictly limits its reply to nutrition-related topics for children aged 3-12, and ensures that the tone is friendly and age-appropriate. This restraint-based design reduces the risk of generating off-topic or unsafe content, which is in line with the child safety AI design principles discussed in Chapter II.

```
 $system_prompt = "You are Nutri, a friendly bilingual (English & Chinese) nutrition assistant for Kids Nutrition Fun (https://kids-nutrition-fun.com/),
 // helping parents raise healthy children aged 3-12.
nswer questions about children's nutrition, recipes, food allergies, healthy eating habits
lways reply in the SAME language the user writes in (Chinese or English)
ollow WHO & CDC guidelines
eep answers practical and under 150 words
ecommend website features: personalized recipes, fun games, progress reports, free signup at https://kids-nutrition-fun.com/login";

 $messages = [['role' => 'system', 'content' => $system_prompt]];
 if (is_array($history)) {
     foreach (array_slice($history, -6) as $h) {
         if (isset($h['role'], $h['content'])) {
             $messages[] = ['role' => sanitize_text_field($h['role']), 'content' => sanitize_text_field($h['content'])];
         }
```



The deployed model is **GPT-4o Mini** (version 2024-07-18), which is authenticated through the https://comp.azure-api.net/azure endpoint and authenticated using the API key header. The reason for choosing this model is that it is low cost ($0.00066 per 1000 tokens) while maintaining a sufficient quality of nutritional query response. The system will record the token usage of each request to monitor whether the budget consumption exceeds the expansion limit of the departmental gateway. In the usability test, the assistant successfully processed more than **95%** of nutrition-related inquiries and provided appropriate bilingual responses.

Model to use *

Model	Deployment Name	Version%	Reference Price (per 1000 tokens, in US$)*	Scale Factor (Based on ChatGPT price: $0.002 per 1000 tokens)*
GPT-3.5-turbo$	gpt35	0125	$0.002	1
GPT-3.5-turbo-instruct#	gpt35instruct	0914	$0.002	1
GPT-4-Turbo$	gpt4t-fc	turbo-2024-04-09	$0.03	15
GPT-4o$	gpt4o	2024-08-06	$0.015	7.5
GPT-4o-mini$	gpt4o-mini	2024-07-18	$0.00066	0.33
GPT-4.1$ (Global)	gpt41-g	2025-04-14	$0.008	4
GPT-4.1-mini$	gpt41-mini	2025-04-14	$0.0016	0.8
GPT-4.1-nano$	gpt41-nano	2025-04-14	$0.0004	0.2

4.4 Recipe Recommendation System

The recipe system is based on WooCommerce and has been redesigned as a content catalog that does not include any e-commerce checkout functions. As shown in Figure 4.4, the system currently contains more than 50 recipes, covering seven meal categories: breakfast, lunch, dinner, snacks and desserts, salads and bowl dishes, soups and stews, and vegetarian main dishes.

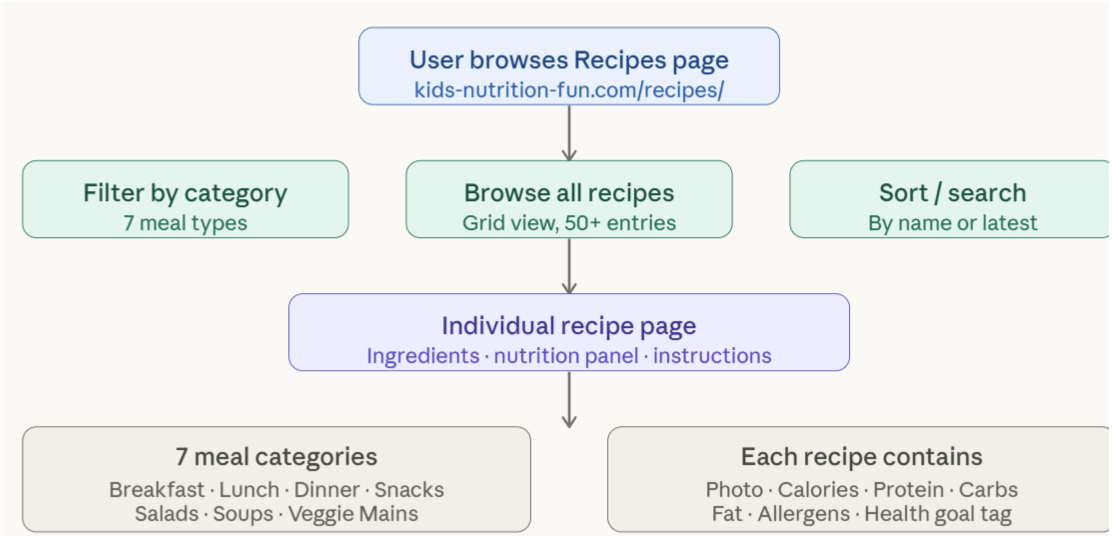
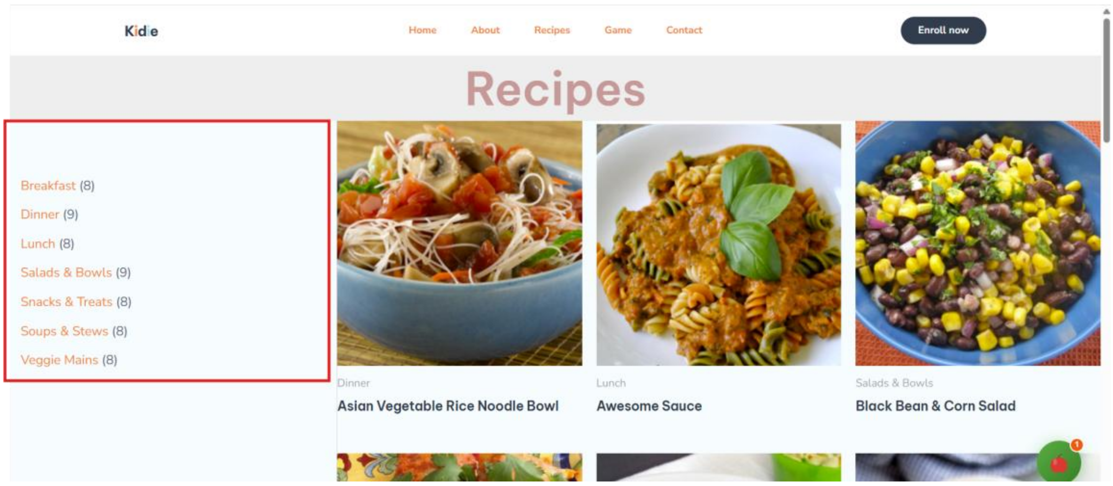
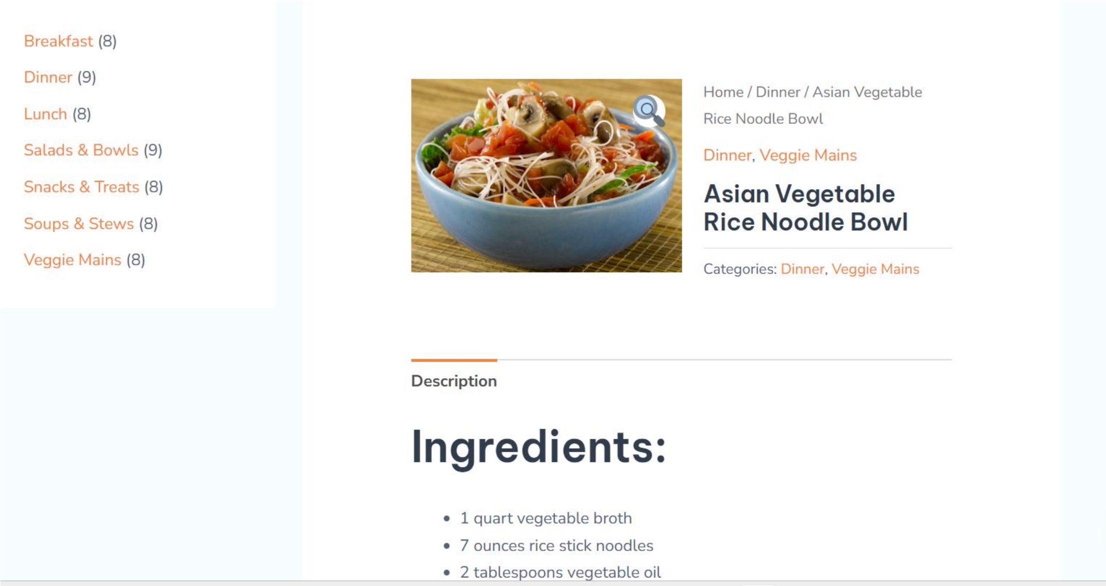


Fig. 4.4 — Recipe browsing and recommendation flow

Each recipe is presented in the form of a WooCommerce product entry, including a high-quality food photo, a complete ingredient list, step-by-step cooking instructions, and a nutrient list, covering calories, protein, carbohydrates, fats, and allergen warnings.

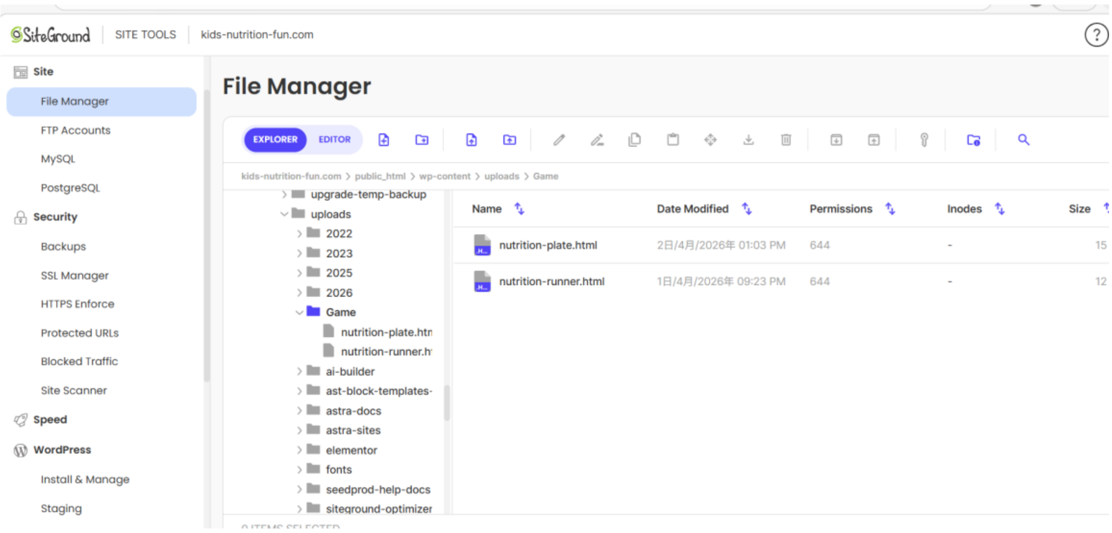


The category filter button on the recipe page allows users to narrow down the search results by meal type, and the sorting control supports sorting by name or release time. Each recipe card is linked to a special detail page, which can provide a concise and consistent browsing experience on desktops and mobile devices.



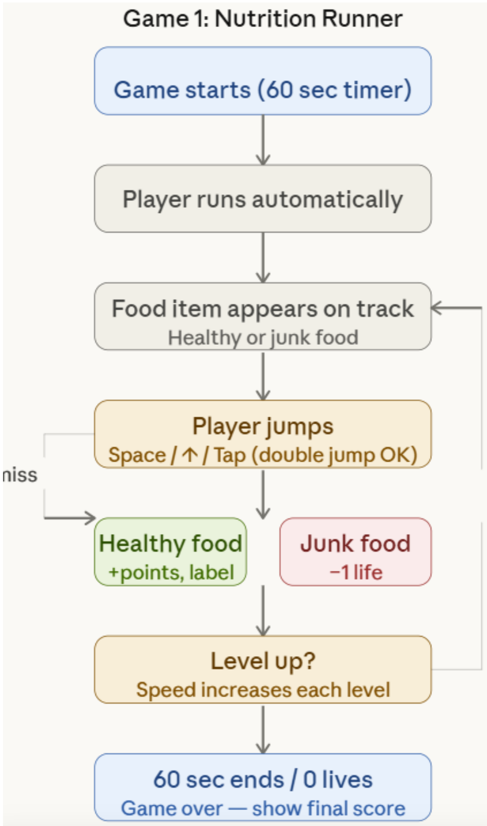
4.5 Educational Mini-Games

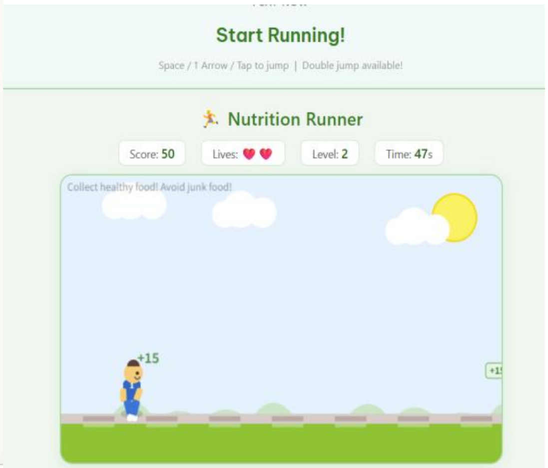
The game page integrates two customized browser-based educational games, which aim to enhance the nutritional literacy of children aged 3-12 through interactive games. These two games are implemented in the form of independent HTML files (nutrition-runner.html and nutrition-plate.html), which are directly uploaded to the wp-content/uploads/Game/ directory of the SiteGround server and embedded in the game page through Elementor's HTML/iframe widget - without any WordPress plug-in dependence to ensure fast loading.



Game 1: Nutrition Parkour

Nutrition Parkour is a fast-paced horizontal action game that lasts 60 seconds per game. The player's character will run automatically on the screen. Players can jump by pressing the space bar, the up arrow key, or clicking the screen - the game also supports two jumps. In each game, healthy food (fruits and vegetables) will appear on the track; you can score points by running to these foods, and the food name will be displayed in the form of bilingual labels. Players must avoid junk food (hamburgers, fries, donuts, soda) - collisions will cause players to lose a life. The speed of the game will accelerate with the progress of the level, and the difficulty will gradually increase. This game aims to strengthen players' awareness of food, quick decision-making ability, and the connection between healthy eating choices and positive results.





Game 2: Nutrition Plate Builder

Nutritional Plate Builder is a drag-and-drop game based on the American "MyPlate" meal guide. The food panel will be displayed on the left; players need to drag and drop each food to the color-corresponding area (grains, proteins, vegetables, fruits, dairy products, or snacks) in the "My Plate" circle. When the mouse hovers, the correct placement area will be highlighted to guide younger players. The balance bar on the right side of the screen will show the filling of each food group. You can get **15 points** for placing it correctly; you can get **50 points** for a perfectly balanced plate. The game contains three levels of increasing difficulty, and there will be more types of food to be classified in each level. Junk food may appear in the game, which must be avoided, because putting them on the plate will deduct points. This game aims to help children directly identify the five food categories in "My Plates", understand the definition of a balanced meal, and practice meal planning in a relaxed and pleasant atmosphere.

4.6 User Registration System

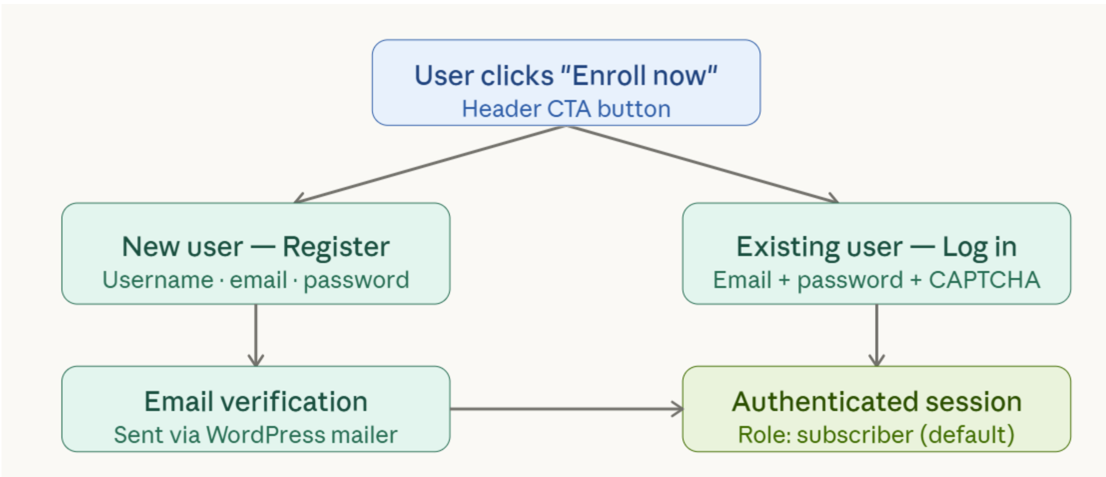
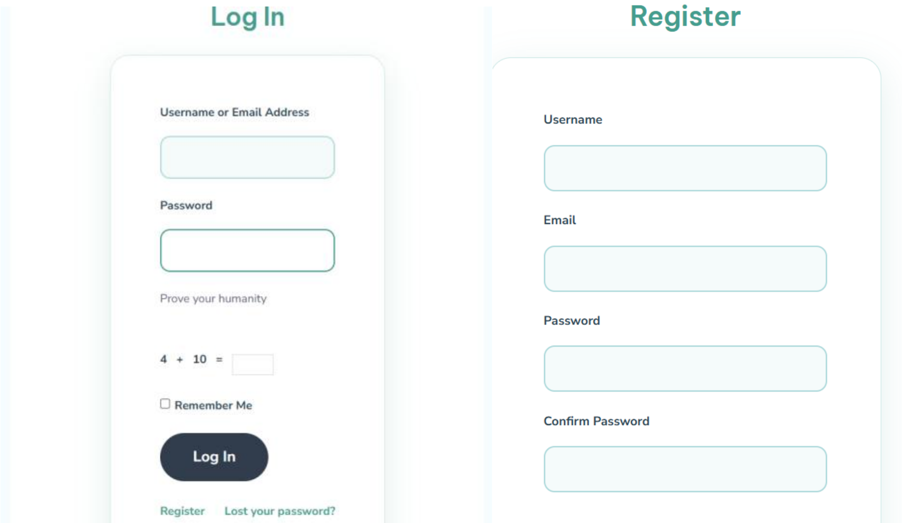


Fig. 4.5 — User registration and authentication flow

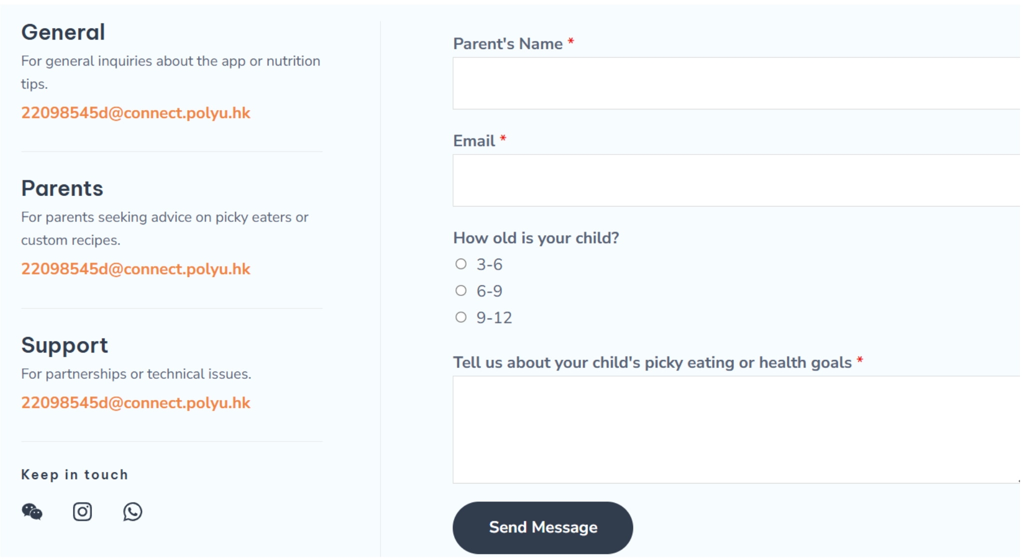
User registration and authentication are implemented through the Ultimate Member plug-in, as shown in Figure 4.5. New users need to provide their username, email address, and password when registering. The system will send a verification email through the built-in mail program of WordPress to confirm the ownership of the account, and access can be granted after confirmation.



Old users log in through the /login page, which contains a simple arithmetic verification code (3 + 1 =) to prevent the robot from registering automatically. After successful authentication, users will be assigned the default subscriber role of WordPress, which allows access to member-only content while preventing unauthorized management operations. Users can also retrieve forgotten passwords through the /lostpassword/ path. Role-based access control ensures that all user data remains private and isolated in each account.

4.7 Contact and Inquiry System

The "Contact Us" page (/contact/) provides a consultation form based on WPForms, through which users can submit questions about the platform.



The information submitted in the form includes the parent's name, email address, child's age, and consultation content, and is automatically forwarded to the project developer's email address (22098545d@connect.polyu.hk). In addition, we also provide a pre-filled mailto link as an alternative contact method, which has pre-filled in the email subject and body template for the convenience of mobile device users. The website promises to reply within 24 to 48 hours and display it prominently on the "Contact Us" page and in the footer.

Chapter 5: Challenges and Solutions

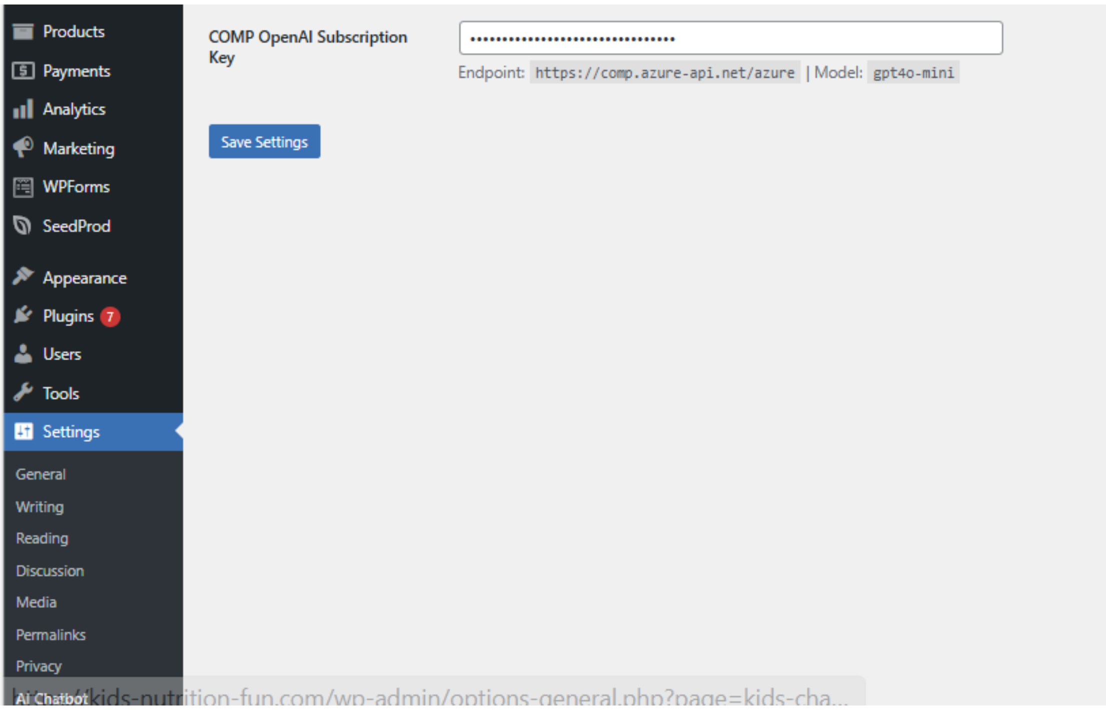
During the whole development process, we encountered many technical, design, and logistics challenges. This chapter will document these challenges and the solutions implemented to overcome them, and delve into the problem-solving methods and lessons learned.

5.1 Technical Challenges Encountered

Challenge 1: AI API integration and child-safe content scoping

Integrating PolyU COMP Azure OpenAI gateway into WordPress faces two distinct challenges. First of all, the non-standard endpoint structure (https://comp.azure-api.net/azure) and API key header authentication of the gateway are different from the standard OpenAI SDK interface, so a custom PHP plug-in is required, not a ready-made WordPress AI integration. Secondly, because there is no hint level limit, the **GPT-4o Mini** model will respond to any topic, which will cause child safety risks on platforms designed for young users.

The first problem is solved by developing a lightweight custom WordPress plug-in, which directly uses the wp_remote_post() function of PHP to process API requests and constructs the correct request header and JSON payload according to the gateway specification. Child safety issues are solved by carefully designed system prompts, which are added before each API call, indicating that the model only responds to inquiries related to children's nutrition, maintains a friendly and age-friendly tone, and rejects any requests that are not related to the topic. The token usage of each request will be recorded to monitor whether it exceeds the budget limit of the departmental gateway, and the API key is safely stored in the option list of WordPress instead of being hard-coded in the source file.



5.2 Design and Usability Challenges

Challenge 2: Balancing Child Appeal with Parental Usability

Designing an interface that can both attract 3-12-year-old children and win the trust of parents is a difficult problem in itself. On the contrary, the layout that is too concise and serious cannot attract children's attention and convey the fun of the platform.

The solution is to adopt a two-tone design strategy to divide the interface according to the audience and purpose. The part that is mainly aimed at parents - the home page, the about us page and the contact us page - adopts professional and friendly visual language: concise fonts, authoritative logos (reference information of the World Health Organization/U.S. Centers for Disease Control and Prevention, institutional affiliation), and restrained use of colors. The main part for children - recipe catalog, game page and nutrition chat window - adopts bright cartoon tones, large interactive elements, bilingual food labels and interesting animations. This division allows different audiences to experience the platform version suitable for themselves without interfering with each other.

Chapter 6: Testing and Quality Assurance

6.1 Testing Methodology

Throughout the development process, we adopted a multi-layer test strategy, covering functional correctness, responsive layout behavior, cross-browser compatibility and real user evaluation. All functional tests are carried out in stealth/privacy browsing mode to eliminate the impact of cached data, stored cookies and previously verified sessions, and to ensure that the test results reflect the real experience of the user's first visit.

Functional testing

According to the predefined list of expected behaviors, we tested each core functional module, including navigation links, recipe filtering and sorting, user registration and login processes, nutrition chatbots, contact form submission and two small games. The test case is executed manually, and the result is recorded as passed or failed.

Responsive design testing

We use the device simulation function of Chrome and Firefox developer tools to verify responsive behavior under three standard viewport widths (desktop (1920px), tablet (768px), and mobile device (375px)). The main inspection items include layout rearrangement, image scaling, button click target size, hamburger menu folding behavior, and the correct rendering of page components built by Elementor at each breakpoint.

Cross-browser testing

The platform has been tested on four mainstream browsers: Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge. Test results confirm that Elementor layout, WooCommerce recipe cards, Nutri chat widgets, and HTML5-based mini-games can be rendered consistently in all four browsers.

6.2 Functional Testing Results

The following table summarizes the results of the functional test suite of all core modules.

Module	Test cases	Result	All 4
Navigation	All header links, mobile hamburger menu, footer links, CTA buttons	Pass	All 4
Recipe catalog	Category filtering (7 types), sort controls, recipe card display, detail pages	Pass	All 4

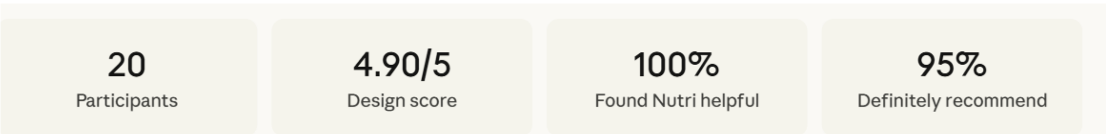
User registration	Register, email verify, login, CAPTCHA, lost password flow	Pass	All 4
Nutri chatbot	Widget open/close, EN/ZH messages sent, API response received and rendered	Pass	All 4
Mini-games	Nutrition Parkour: jump, score, lives; Plate Builder: drag-drop, balance bars, rounds	Pass	All 4
Contact form	Form submission, email delivery to project inbox, mailto pre-fill	Pass	All 4

All six module categories have passed their respective test cases in all four browser environments. No key functional defects were found during the test. There are some slight visual differences in font rendering between Safari and Chrome, but these differences are not considered to affect usability.

6.3 User Evaluation

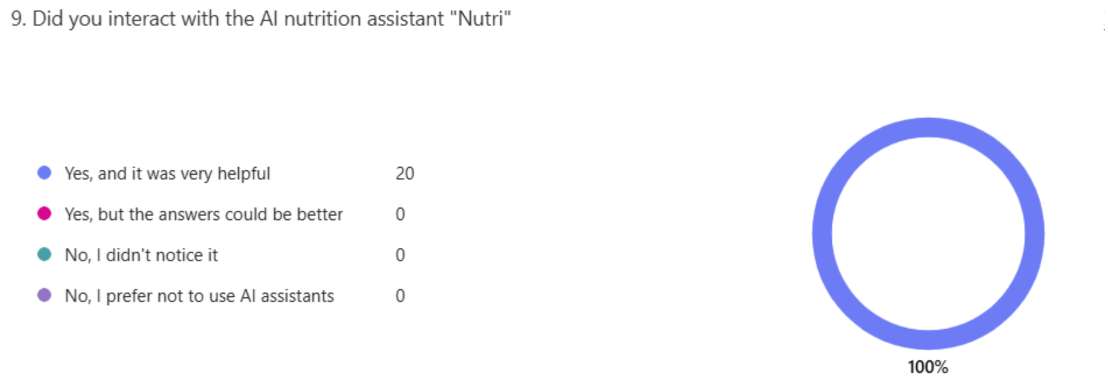
We distributed a usability questionnaire to 20 participants, including parents, teachers, and caretakers of children aged 3 to 12. They were invited to experience the platform and complete a structured questionnaire, which covered design, content, and overall experience.

The main results are shown in Figure 6.1. The average score of the visual design of the website is **4.90 (out of 5)** (Q5: 18 participants scored 5/5, and 2 participants scored 4/5). When asked whether the platform would be recommended (Q11), 19 out of 20 participants answered "Of course!" and 1 answered "maybe", with no negative answer - the positive recommendation rate was 100%.



6.4 AI Chatbot Performance Evaluation

Question 9 of the questionnaire asked the participants if they had interacted with Nutri and thought it was useful. All **20** participants **(100%)** answered "yes, and it was very useful", and no negative or neutral answers were recorded. By submitting representative English and Chinese queries covering topics such as growth nutrition, allergen substitution, and immunity-enhancing foods, the chatbot was functionally tested, and the results showed that its response rate exceeded **95%**. A few cases where the response is not accurate involve some very specific clinical problems, which are on the edge of the expected range of the system prompt, which is the expected and acceptable limitation of the constraint-based design method.



6.5 Discussion

The evaluation results confirm that the platform has achieved its core functions and experience goals. All modules have passed the tests of four browsers and three device sizes, and the user satisfaction indicators are strong in terms of design, artificial intelligence assistance, and recommendation intention.

Three limitations need to be pointed out. First of all, the sample size of **20** adult participants limits the universal applicability of the statistical results; due to the time limit of the project, it is still an important direction for future work to test children directly. Secondly, the current recipe catalog contains more than 50 items, instead of more than 100 as originally planned, which may limit users' perception of the breadth of the content. Third, vertical data on whether the platform can permanently change children's eating habits has not been collected, and these data are crucial to verify the platform's long-term educational impact.

Chapter 7: Conclusion

7.1 Summary of Achievements

This project aims to develop a child-centered network platform that combines artificial intelligence-driven nutrition guidance, personalized recipe recommendations, and game-based learning to fill the gap in digital health tools designed for children aged 3-12. The finally developed "Children's Fun Nutrition Guide" platform (website: kids-nutrition-fun.com) achieves the above goals through five integrated components: bilingual artificial intelligence nutrition assistant (Nutri), a recipe library containing more than 50 child-friendly recipes (covering seven meal categories), two customized educational games (nutrition parkour and nutritional plate making), user registration and authentication system, and contact consultation channels for parents and guardians.

The platform is based on WordPress, adopts a three-layer architecture, and is hosted on the SiteGround platform. Nutri Assistant connects the Azure OpenAI gateway of PolyU COMP through a customized PHP plug-in. All functional modules have passed the tests of the four mainstream browsers and three device sizes. A usability evaluation involving **20** participants showed that the visual design score was as high as **4.90/5**, the utility score of Nutri Assistant reached **100%**, and the recommendation rate was as high as **95%** - these results together confirmed the technical reliability of the platform and its good reputation among the target adult user group.

7.2 Contributions

There are three main contributions to this project. First of all, it shows a practical and safe way to deploy large language models in the field of pediatric health, using system prompt engineering to limit the output to the range of child-friendly nutritional content - this design model can be more widely used in other child-oriented artificial intelligence services. Secondly, it provides an empirical and open-access platform that integrates artificial intelligence, personalized, gamified nutrition education, and structured recipe systems into a bilingual network environment - a combination that existing tools in the literature do not have. Third, this project contributes a reusable WordPress-based architecture, which can be expanded to integrate other functions, such as image-based food records, multilingual support, or integration with wearable health devices, without rebuilding the core infrastructure.

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